Weixiang Fang, PhD

Department of Biomedical Engineering

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Research Focus

My research asks how cells decide their fates, and how what model organisms reveal about these decisions can be carried into human. I develop **computational methods** for data from **emerging genomics technologies**, building models of how cells divide, change state and respond to signals, learned from **genomic records**, the heritable edits a recorder writes into a cell's own DNA.

Research program

- **Decoding genomic records at scale.** Probabilistic models in which editing and silencing depend on cell type and developmental time, estimated jointly with the lineage tree, so that a conditional recorder read alongside a constitutive one yields a **decorated lineage tree**, which in addition to the divisions reveals the signals and cell state changes along its branches. A transformer proposes tree rearrangements, bringing likelihood-based reconstruction to millions of cells.
- **A generative model of development in lineage and space, to learn intrinsic and extrinsic control of cell fate.** A model in which components learned from data capture how cells divide, change state, move and respond to signals, fitted jointly to lineage, time-course and spatial data. In silico transplantation maps each state's extrinsic plasticity, the share of its fate that signals can still change, and the signals that could redirect it.
- **Cross-species transfer learning for the regulatory control of fate decisions.** A representation of regulatory activity across species and developmental stage, with human and mouse cell states in one embedding. Swapping sequence and species names the human element behind an animal fate decision, and a knowledge base transfers decades of animal evidence into testable hypotheses in human.

Selected Works

- **Quantitative fate mapping: A general framework for analyzing progenitor state dynamics via retrospective lineage barcoding**. *Cell*, 2022.
- **Quantifying Functional Conservation of Human and Mouse Regulatory Elements via FUNCODE**. *bioRxiv preprint*, 2024. (In revision *Genome Biology*)
- **A branching process model of cell lineage links positional signaling to the canalization of clonal fate**. *Manuscript in preparation*, 2026.

Academic Appointments

Postdoctoral Fellow in Biomedical Engineering 2022–Present
Johns Hopkins University · Baltimore, MD · PI: **Dr. Reza Kalhor**

Education

PhD in Biostatistics 2016–2022
Johns Hopkins University · Baltimore, MD
Dissertation: "**Quantitative Methods for Genomics and Lineage Tracing Data**" · Advisor: **Dr. Hongkai Ji**

Master's program in Biostatistics 2014–2016
Johns Hopkins University · Baltimore, MD · Transferred to PhD program

B.Eng in Engineering Science 2010–2014
Stony Brook University · Stony Brook, NY

Major in Psychology 2009–2010
Fudan University · Shanghai, China · Transferred to Stony Brook University

Publications

Peer-reviewed journal articles

1. **W. Fang**, C. M. Bell, A. Sapirstein, S. Asami, K. Leeper, D. J. Zack, H. Ji, and R. Kalhor, 2022. "Quantitative fate mapping: A general framework for analyzing progenitor state dynamics via retrospective lineage barcoding". *Cell*. doi.org/10.1016/j.cell.2022.10.028
2. W. Zhou, Z. Ji, **W. Fang**, and H. Ji, 2019. "Global prediction of chromatin accessibility using small-cell-number and single-cell RNA-seq". *Nucleic Acids Research*. doi.org/10.1093/nar/gkz716
3. X. Chamling, A. Kallman, **W. Fang**, C. A. Berlinicke, J. L. Mertz, P. Devkota, I. E. M. Pantoja, M. D. Smith, Z. Ji, C. Chang, A. Kaushik, L. Chen, K. A. Whartenby, P. A. Calabresi, H.-Q. Mao, H. Ji, T.-H. Wang, and D. J. Zack, 2021. "Single-cell transcriptomic reveals molecular diversity and developmental heterogeneity of human stem cell-derived oligodendrocyte lineage cells". *Nature Communications*. doi.org/10.1038/s41467-021-20892-3
4. J. S. Handler, Z. Li, R. K. Dveirin, **W. Fang**, H. Goodarzi, E. J. Fertig, and R. Kalhor, 2025. "A cell-state axis underlying colonization in carcinomas with implications for metastasis risk prediction and interception". *Cell Reports*. doi.org/10.1016/j.celrep.2025.116701
5. W. Li, C. Berlinicke, Y. Huang, S. Giera, A. G. McGrath, **W. Fang**, C. Chen, F. Takaesu, X. Chang, Y. Duan, D. Kumar, C. Chang, H.-Q. Mao, G. Sheng, J. C. Dodge, H. Ji, S. Madden, D. J. Zack, and X. Chamling, 2023. "High-throughput screening for myelination promoting compounds using human stem cell-derived oligodendrocyte progenitor cells". *iScience*. doi.org/10.1016/j.isci.2023.106156
6. C. J. Keuthan, J. A. Schaub, M. Wei, **W. Fang**, S. Quillen, E. Kimball, T. V. Johnson, H. Ji, D. J. Zack, and H. A. Quigley, 2023. "Regional gene expression in the retina, optic nerve head, and optic nerve of mice with optic nerve crush and experimental glaucoma". *International Journal of Molecular Sciences*. doi.org/10.3390/ijms241813719

Preprints

7. **W. Fang**, C. Chen, B. Zhang, Y. Wang, R. Zhao, W. Zhou, and H. Ji, 2024. "Quantifying Functional Conservation of Human and Mouse Regulatory Elements via FUNCODE". *Preprint; in revision at Genome Biology*. doi.org/10.1101/2024.10.31.620766
8. T. Fabiha, I. Evergreen, S. Kundu, A. Pampari, S. Abramov, A. Boytsov, K. Strouse, K. Dura, **W. Fang**, G. Kerner, J. Butts, T. Ali, A. Gschwind, K. S. Mualim, J. E. Moore, Z. Weng, J. Ulirsch, H. E. Ji, J. Vierstra, T. E. Reddy, S. B. Montgomery, J. Engreitz, A. Kundaje, R. Tewhey, A. Price, and K. Dey, 2024. "A consensus variant-to-function score to functionally prioritize variants for disease". *Preprint*. doi.org/10.1101/2024.11.07.622307

Submitted

9. J. E. Forsmo, **W. Fang**, J. D. Lin, M. Vazquez-Arroyo, L. Fang, J. Jun, W. Lin, and R. Kalhor, 2026. "Genomic recording of gene-expression history along lineage trees". *Submitted*

In preparation

10. **W. Fang**, Y. Yang, J. D. Lin, and R. Kalhor, 2026. "A branching process model of cell lineage links positional signaling to the canalization of clonal fate". *Manuscript in preparation*

Book chapter

11. **W. Fang**, Y. Yang, H. Ji, and R. Kalhor, 2025. "Reconstructing Progenitor State Hierarchy and Dynamics Using Lineage Barcoding Data". *Lineage Tracing: Methods and Protocols*. doi.org/10.1007/978-1-0716-4310-5_9

Software

- **QFM** (R package). Quantitative fate mapping from lineage barcodes. **Phylotime** reconstructs time-scaled cell phylogenies by maximum likelihood under a general model of barcode mutagenesis, and **ICE-FASE** infers the progenitor state hierarchy, commitment times, population sizes and commitment biases from them. Code at github.com/Kalhor-Lab/QFM, tutorials, archived at Zenodo.
- **FUNCODE** (resource). Genome-wide functional conservation scores for human and mouse regulatory elements. Code at github.com/wefang/funcode; scores on the ENCODE portal and as UCSC Genome Browser track hubs for hg38 and mm10.

Research Experience

Postdoctoral research

Advisor: Dr. Reza Kalhor 2022–Present

- Developed **Phylotime** and **ICE-FASE**, under the **Quantitative Fate Mapping** framework (*Cell*, 2022), reconstructing dated lineage trees and the hidden progenitor states and timing of fate decisions from **lineage barcodes**.
- Developed the method that reconstructed the **decorated lineage tree** of the ihgRNA conditional recorder, recovering the history of Pax6 expression across the developing retina (submitted).
- Building scalable **lineage tree reconstruction** from engineered and somatic barcodes, including **PhyloTime+**, a C library for whole-tree optimization.
- Building **LINST**, a generative model of development linking a cell's state and the extrinsic signals it receives to the fates of its descendants (manuscript in preparation).

Doctoral research

Advisor: Dr. Hongkai Ji 2014–2022

- **FUNCODE**: genome-wide scores for **functional conservation of regulatory DNA** between human and mouse, built on the ENCODE DNase, ATAC and histone ChIP compendium and released as UCSC track hubs for hg38 and mm10.
- **Variational autoencoder** for prediction of regulome signal using RNA-seq with extension to **single cell RNA-seq**.
- **Statistical learning of local patterns of cis-regulation**: a hierarchical mixture model for the joint activity state of regulatory elements within a locus, fitted to ENCODE DNase hypersensitivity across 123 cell types within Hi-C topological domains, and applied to denoise low-input and single-cell **ATAC-seq**.
- **Functional data analysis of regulatory activity across cell types**: penalized spline models fitted along a cross-species ordering of ENCODE and Roadmap samples, validated on 11,938 orthologous gene pairs, the approach that became the conservation scores in FUNCODE.

Undergraduate research

Advisor: Dr. Juhyuk Moon, Multi-Scale Structural Materials Lab 2013–2014

- Resolved the crystal structure of **yeelimite** ($Ca_4Al_6O_{12}SO_4$), the principal phase of calcium sulfoaluminate cement, by **first-principles DFT** energy minimization across competing cubic, tetragonal and orthorhombic space groups reported in the diffraction literature
- Built **thermodynamic models** of Portland cement hydration by Gibbs energy minimization (GEMS, CEMDATA), covering phase assemblage, pore solution chemistry and temperature dependence in ternary blends with limestone and fly ash
- Measured the thermal behavior of elemental sulfur and **sulfur polymer concrete** by **differential scanning calorimetry**, the material's main application being hazardous waste stabilization

Presentations

Invited talks

- "A scalable approach for characterizing functional conservation of regulatory elements between human and mouse". **ENCODE Consortium Meeting 2021**, comparative genomics section. [Online due to COVID-19 pandemic]
- "Mapping complex cell fate dynamics using cell phylogeny". **2022 4DN annual meeting**, San Diego, 2022
- "Mapping complex cell fate dynamics using cell phylogeny". **4DN Scientific Webinar Series**, Delivered Online, July 31, 2023

Poster presentations

- "Single cell lineage model bridges cell fate and state in development". **CSHL Biological Data Science**, Cold Spring Harbor, NY, November 2024.
- "Understanding developmental variabilities by lineage-resolved single cell model". **EMBO Workshop: Lineage tracing: Dynamics, cellular memory, and somatic evolution**, Sant Feliu de Guíxols, Spain, September 2025. [Delivered remotely]
- "Understanding developmental variabilities by lineage-resolved single cell model". **2026 Lake Conference: Modeling Life from Cells to Tissues: Emergence and Self-Organization**, Allen Institute, Seattle, WA, April 2026.

Presentations by co-authors

- Cell lineages across scales, space and time. Short Talk: "A general framework for inferring progenitor state hierarchy and dynamics from retrospective lineage barcoding" * Presented by **Reza Kalhor**
- Epigenome isoform analysis with applications. JSM, Chicago, IL, Aug 3, 2016 *Presented by **Hongkai Ji**

Honors and Awards

Outstanding Performance on the Biostatistics PhD Comprehensive Exam 2017
Johns Hopkins University

Kocherlakota Award for Best Performance on the Biostatistics Comprehensive Examination by a Master's Student 2015
Johns Hopkins University

Academic Excellence and Outstanding Achievement Award 2012–2013
Stony Brook University

Fudan University First-class Renmin Scholarship 2010
Fudan University

Affiliations

- **The Encyclopedia of DNA Elements (ENCODE) Consortium**, leading analysis on comparative functional genomics
- **4D Nucleome (4DN) program**

Mentoring Experience

Yi Yang · Master's student in Biomedical Engineering 2022–2024
Johns Hopkins University · Now **PhD candidate at Johns Hopkins Biomedical Engineering**
Research projects: Dynamics inference from time-course cell state composition data; web interface for exploring cell phylogeny

Smriti Srikanth · Undergraduate student in Biomedical Engineering 2022–2024
Johns Hopkins University · Now **Computation Associate at Broad Institute**
Research projects: Mosaic fraction analysis to analyze bulk lineage tracing data; model for simulating gene expression on single cell phylogeny in development

Ashwin Rajendran · Master's student in Biomedical Engineering 2023–2025
Johns Hopkins University · Now **PhD candidate at Florida State University**
Research project: Scalable phylogeny inference using diploid lineage barcodes with heritable and random missing data

Teaching Experience

Teaching Assistant 2015–2021
Johns Hopkins University · Department of Biostatistics, Baltimore, MD
For graduate students in biostatistics and related quantitative fields

- **JHSPH 140.688** Statistics for Genomics (Spring 2020)
- **JHSPH 140.776** Statistical Computing (Fall 2017, Fall 2018)

For public health and laboratory scientists

- **JHSPH 140.621–624** Statistical Methods in Public Health (Fall 2016, Spring 2017, Fall 2017, Spring 2018, Spring 2020, Fall 2020, Spring 2021)
- **JHSPH 140.615** Statistics for Laboratory Scientists (Spring 2016, Spring 2019; wrote problems and laboratory exercises with the course director)
- **AS.280.345** Public Health Biostatistics (Fall 2015, Fall 2019)

Undergraduate Teaching Assistant 2013–2014
Stony Brook University · Stony Brook, NY

- Fundamentals of Engineering Chemistry
- Engineering Laboratory

Technical Skills

Statistics. **Likelihood-based inference** for stochastic processes and **phylogenetics**, **Bayesian inference**, MCMC and probabilistic programming (**Stan**), generalized linear and **mixed-effects models**, **survival analysis**, **functional regression**, **high-dimensional statistics**, experimental design

Deep learning. **PyTorch**, **TensorFlow/Keras**

Languages. **R**, **Python**, **C/C++** (CMake), Bash/shell, SQL

Single cell. **Seurat**, **Scanpy**

Python. NumPy, pandas, SciPy, scikit-learn, matplotlib

R. tidyverse, **Bioconductor**, ggplot2

HPC and cloud. **CUDA** and GPU training, **SLURM** CPU and GPU clusters, **AWS**, **Docker** and **Singularity** containers

Tools. **Git**, LaTeX

Wet Lab Experience

Cell and molecular biology. Cell Culture, Cell Transfection, Cryopreservation, Cell Sorting (FACS), Optimizing DNA Amplification (PCR), NGS Library Preparation

Tissue preparation. Murine Embryo Microdissection, Fresh Frozen Tissue Preparation, Cryosectioning, Laser Capture Microdissection

Imaging. Immunohistochemistry, Fluorescence Microscopy